

Data hierarchy

30 athlete codes (inference units)
6 fixed English action labels
50 archived trials per cell
9,000 paired coordinate files

Archive audit

Collapsed 50 duplicated 400-row blocks
190 body trials contain structural zeros
1,180 nominal lengths exceed 200 frames
Codes 31–40 quarantined

Operational waveforms without event registration

Racket: framewise 3-D centroid displacement
Body: mean displacement across available joints
Observed segment normalized to 200 nodes
Not face angle, centre of mass, or coordination

Contextual decomposition of six fixed labels

Finite-set action contrast dispersion
+ athlete + athlete $\times$ fixed action
+ within-cell unexplained variation (descriptive)

Twelve action-specific design models

6 labels $\times$ 2 representations estimate B and W
 L^2 -trace relative reliability $R(n)$
Trial counts summarized at 0.80 and 0.90

Uncertainty, assumptions, and inferential boundary

Left: 5,000 whole-athlete bootstrap replicates;
continuous thresholds, interval diagnostics, and leave-one-out influence
Right: archive-order trace correlation and AR(1) scenarios;
peak registration, 200-row boundary, balanced resampling, and Hampel-type rule
Boundary: within-session relative distinguishability: no extrapolation
to all techniques, populations, days, or racket sports